

Notes on Erdős Problem #409: the forest structure of $n \mapsto \varphi(n) + 1$

Ankit*

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Abstract

Let $f(n) = \varphi(n) + 1$ and let $F(n)$ be the number of iterations of f needed to reach a prime (Erdős Problem #409, a question of Finucane; Guy B41 [4]; OEIS A039651, A229487 [5]). We prove that for 100% of primes $p \equiv 3 \pmod{4}$, exactly two integers ever reach p under iteration of f , namely p and $2p$; this establishes, for a full-density subfamily of primes, the conjecture recorded on the problem's discussion thread that only finitely many integers reach any fixed prime. We further prove $F(n) \leq 5\sqrt{n} + 1$ unconditionally, and that Dickson's conjecture implies F is unbounded, via explicit admissible systems of linear forms built on the map $q \mapsto (3q + 1)/2$. Exhaustive computations certify that every basin for every prime $p < 30,000$ is finite, and determine $\max_{n \leq 10^7} F(n) = 38$. Finally we recast the question of infinite basins as a multitype branching process in the 2-adic type $v_2(m - 1)$, and measure its Perron root to be ≈ 1.07 – 1.17 across scales 10^4 – 10^7 : marginally supercritical, which places the recorded finiteness conjecture, if true, near criticality. The theorems are elementary and possibly folklore; none is recorded on the problem page as of 20 December 2025.

1 The problem

Erdős Problem #409 [1] asks, for $f(n) = \varphi(n) + 1$:

Q1. How many iterations of f are needed before a prime is reached? (Write $F(n)$ for this count, with $F(p) = 0$ for p prime; OEIS A039651. Recorded remarks: $F(n) = o(n)$ trivially, and $F(n) = 1$ infinitely often (Cambie).)

Q2. Can infinitely many n reach the same prime?

Q3. What is the density of n which reach any fixed prime?

Additionally recorded on the discussion thread (checked 22 July 2026): (i) a conjecture, referenced to OEIS A229487, that only finitely many integers reach any fixed prime—i.e. Q2 conjecturally *no* and Q3 conjecturally *zero*; (ii) explicit constructed examples with $F(n) = 67$ (e.g. $n = 2548851069$), all reaching the prime 9005041. Below: two unconditional theorems—Theorem 2 *proves the recorded finiteness conjecture for a full-density subfamily of primes* ($p \equiv 3 \pmod{4}$)—one conditional theorem on Q1, exact certified computations complementing the constructed records, and a quantitative branching analysis that sits in mild tension with the recorded conjecture.

*Working notes, 22 July 2026; not peer-reviewed. These notes were prepared with substantial AI assistance (Claude, Anthropic); every numerical claim was machine-verified with the accompanying code. All proofs have been verified by hand and all computations independently reproduced by the author.

2 Structure

Write $B(p) = \{n \geq 1 : \text{the } f\text{-trajectory of } n \text{ contains } p\}$, the *basin* of the prime p .

Theorem 1. (i) $f(1) = f(2) = 2$, and $f(n)$ is odd for every $n \geq 3$.

(ii) The fixed points of f are exactly the primes.

(iii) If $n \geq 4$ is composite then $f(n) \leq n - \sqrt{n} + 1 < n$.

(iv) Every trajectory reaches a prime in finitely many steps; hence $\mathbb{N} = \bigsqcup_p B(p)$ (with $1, 2 \in B(2)$).

(v) No even number ≥ 4 is a value of f ; hence in the preimage forest every even number is a leaf.

(vi) If $f(x) = m$ and $x \neq m$, then $x > m$ (each tree grows strictly upward; there are no cycles).

(vii) $F(n) \leq 5\sqrt{n} + 1$ for all $n \geq 1$.

Proof. (i) $\varphi(n)$ is even for $n \geq 3$. (ii) $\varphi(p) + 1 = p$; conversely $\varphi(n) = n - 1$ forces n prime. (iii) The least prime factor q of composite n satisfies $q \leq \sqrt{n}$, so $\varphi(n) \leq n - n/q \leq n - \sqrt{n}$. (iv) Immediate from (ii), (iii). (v) By (i), the only even value of f is $2 = f(1) = f(2)$. (vi) $\varphi(x) = m - 1 \leq x - 1$, with equality iff x is prime, in which case $x = m$. (vii) For composite $m \geq 4$ we have $m - f(m) \geq \sqrt{m} - 1 \geq \frac{1}{2}\sqrt{m}$, so the number of composite iterates lying in $(x/2, x]$ is at most $(x/2)/(\frac{1}{2}\sqrt{x/2}) = \sqrt{2x}$; summing dyadically, $F(n) \leq \sqrt{2n}(1 - 2^{-1/2})^{-1} + O(1) \leq 5\sqrt{n} + 1$. $\square$

Thus the dynamics is a forest: one finite-or-infinite tree per prime, growing strictly upward, with all even numbers as leaves and all branching through odd composite nodes. *Q2 asks exactly whether some tree is infinite.* Since each vertex has finitely many children ($\varphi(x) = v \Rightarrow x \leq 2v^2$), König's lemma shows a tree is infinite iff it contains arbitrarily long chains, i.e. iff F is unbounded on a single basin.

3 The 2-adic gate: Q2 and Q3 for almost all $p \equiv 3 \pmod{4}$

Theorem 2. Let $p > 3$ be prime with $p \equiv 3 \pmod{4}$. Then

$$f^{-1}(p) \setminus \{p\} = \{2p\} \cup \{q^a, 2q^a : q \text{ prime, } q \equiv 3 \pmod{4}, a \geq 2, q^{a-1}(q-1) = p-1\}.$$

Consequently:

(a) If $p-1$ is not of the form $q^{a-1}(q-1)$ with $q \equiv 3 \pmod{4}$ prime and $a \geq 2$ (call such p non-exceptional), then $B(p) = \{p, 2p\}$: exactly two integers in all of $\mathbb{N}$ ever reach p .

(b) The number of exceptional $p \leq x$ is at most $\sqrt{2x} + O(x^{1/3} \log x)$.

(c) Since $\#\{p \leq x : p \equiv 3 \pmod{4}\} \sim x/(2 \log x)$, statement (a) holds for 100% of primes $p \equiv 3 \pmod{4}$. For these primes the answer to Q2 is no, and to Q3 is density zero (indeed the basin is finite, of size 2).

Proof. Suppose $f(n) = p$, i.e. $\varphi(n) = p - 1 \equiv 2 \pmod{4}$. Write $n = 2^b m$ with m odd. Then

$$v_2(\varphi(n)) = \max(b - 1, 0) + \sum_{q|m} v_2(q - 1),$$

where the sum runs over the distinct odd primes dividing m and each term is ≥ 1 . For the total to equal 1: either $m = 1$, $b = 2$ (so $n = 4$, $\varphi(n) = 2$, $p = 3$, excluded), or $m = q^a$ is a prime power with $v_2(q - 1) = 1$ (i.e. $q \equiv 3 \pmod{4}$) and $b \leq 1$. Then $\varphi(n) = q^{a-1}(q - 1) = p - 1$; the case $a = 1$ forces $q = p$, giving $n \in \{p, 2p\}$. This proves the displayed classification. For (a): $2p$ is an even leaf by Theorem 1(v), so the tree stops. For (b): if $a = 2$ then $q(q - 1) = p - 1 \leq x$ gives $q \leq \sqrt{2x}$; for each $a \geq 3$, $q^{a-1} \leq x$ allows at most $x^{1/(a-1)}$ choices of q , and summing over $3 \leq a \leq \log_2 x$ gives $O(x^{1/3} \log x)$. (c) follows since $\sqrt{x} = o(x/\log x)$. $\square$

Certified verification. Among the 1632 primes $3 < p < 30000$ with $p \equiv 3 \pmod{4}$, exactly 1620 have $B(p) = \{p, 2p\}$, and the exceptional set is precisely

$$\{7, 19, 43, 163, 487, 1459, 4423, 6163, 14407, 19183, 22651, 26407\},$$

matching the $q^{a-1}(q - 1) + 1$ characterization exactly (7, 19, 163, 487, 1459 from $q = 3$; 43, 14407 from $q = 7$; 4423 from $q = 67$; etc.).

Remark. Theorem 2 does *not* resolve Q2/Q3 as posed (which quantify over all primes); it resolves them for a full-density subfamily, concentrating the open content on $p \equiv 1 \pmod{4}$ and the sparse exceptional primes.

4 Exact basins (certificates)

Using an inverse-totient solver validated exhaustively (all $v \leq 2000$ against brute force over $n \leq 8 \cdot 10^6$, justified by the bound $n \leq 2\varphi(n)^2$, itself machine-checked on the range), *every basin $B(p)$ for every prime $p < 30,000$ was computed exactly and is finite* (3245 trees; zero truncations at caps of $4 \cdot 10^4$ nodes and 10^{12} values). Samples:

p	$B(p)$
2	$\{1, 2\}$
3	$\{3, 4, 6\}$
5	$\{5, 8, 10, 12\}$
7	$\{7, 9, 14, 15, 16, 18, 20, 24, 30\}$
11	$\{11, 22\}$
13	31 elements, height 4, max 138
61	57 elements, height 8, max 4314

The largest tree found is at $p = 21089$: $|B| = 5557$, height 17, maximal element 4,431,606. The tallest is at $p = 23509$ (height 20), whose deepest chain, machine-verified link by link, is

$$\begin{aligned} 115243 \rightarrow 108449 \rightarrow 98581 \rightarrow 84493 \rightarrow 80029 \rightarrow 79421 \rightarrow 77533 \rightarrow 74141 \rightarrow 73501 \rightarrow 71101 \\ \rightarrow 70273 \rightarrow 60229 \rightarrow 53761 \rightarrow 52273 \rightarrow 48241 \rightarrow 45685 \rightarrow 36545 \rightarrow 29233 \rightarrow 26401 \\ \rightarrow 24833 \rightarrow 23509. \end{aligned}$$

5 Q1: a conditional unboundedness theorem, and data

Theorem 3 (Dickson $\Rightarrow F$ unbounded). *Fix $L \geq 1$. For integers $t \geq 1$ with $2^L t \neq 4$ define*

$$q_i(t) = 3^{i-1} 2^{L-i+1} t - 1 \quad (1 \leq i \leq L), \quad p(t) = 2^{L+1} t - 3.$$

These $L + 1$ linear forms constitute an admissible system. If $q_1, \dots, q_L, p$ are simultaneously prime, then with $m_i := 3q_i$ one has $f(m_{i+1}) = m_i$ and $f(m_1) = p$, hence $F(3q_L) = L$ exactly. Under Dickson's conjecture [2] this occurs for infinitely many t , for every L ; hence F is unbounded.

Proof. By construction $q_{i+1} = (3q_i + 1)/2$, so $\varphi(3q_{i+1}) = 2(q_{i+1} - 1) = 3q_i - 1 = m_i - 1$, using $\gcd(3, q_{i+1}) = 1$ (guaranteed by $2^L t \neq 4$, which forces every $q_i > 3$); and $\varphi(3q_1) = 2(q_1 - 1) = 2^{L+1} t - 4 = p - 1$. Each m_i is composite, so the trajectory of $3q_L$ takes exactly L steps. Admissibility: all forms are odd; mod 3, choose $t \equiv 2 \pmod{3}$ for L even and $t \equiv 1 \pmod{3}$ for L odd; mod $r \geq 5$, taking $t \equiv 0$ makes every $q_i \equiv -1$ and $p \equiv -3 \not\equiv 0$. $\square$

Verified witnesses (each trajectory literally iterated and checked): $L = 3$: $t = 1$, $F(51) = 3$, landing on 13. $L = 4$: $t = 2$, $F(321) = 4$, landing on 61. $L = 5$: $t = 2905$, $F(1,411,827) = 5$, landing on 185,917. $L = 6$: $t = 750155$, $F(1,093,725,987) = 6$, landing on 96,019,837.

Data. Exact computation of $F(n)$ for all $n \leq 10^7$ gives the record value $F = 38$ at $n = 7,754,613$ (terminal prime 30689), and

$$\max_{n \leq 10^k} F(n) = 4, 7, 11, 18, 28, 38 \quad (k = 2, \dots, 7),$$

strikingly linear in $\log n$; the distribution of F has a geometric tail of ratio ≈ 0.76 per step. The bound of Theorem 1(vii) already improves the recorded “trivially $o(n)$ ”, but the truth appears to be logarithmic:

Conjecture 1. $\max_{n \leq x} F(n) \asymp \log x$. (*Lower-bound mechanism: the chains of Theorem 3 have top element $\approx 2 \cdot 3^L t$; an effective form of Dickson/Bateman–Horn with polynomially bounded least solutions would give $\max_{n \leq x} F(n) \gg \log x$.)*

6 Q2 as a multitype branching process

By Theorem 1, a tree grows only through odd composite nodes m , whose children are the odd solutions x of $\varphi(x) = m - 1$. By the valuation identity in the proof of Theorem 2, the fertility of m is governed by its *type* $t = v_2(m - 1)$: type-1 parents ($m \equiv 3 \pmod{4}$) admit only prime-power children (rare), while higher types branch freely.

Measured mean-offspring matrices $M[t][t']$ (random odd m of size $\sim x$, types capped at 6) have Perron roots

$$\lambda \approx 1.10 \ (x = 10^4), \quad 1.17 \ (10^5), \quad 1.09 \ (10^6), \quad 1.17 \ (3 \cdot 10^6), \quad 1.07 \ (10^7),$$

with seed-to-seed noise ± 0.1 (offspring counts are heavy-tailed; totient multiplicities are unbounded, cf. [3]). The matrix has a telling shape: fertile high-type parents pour most of their children into the sterile type-1 class.

So the process is *marginally supercritical* at every measured scale—yet all 3245 real trees in range go extinct. These are compatible: with $\lambda \approx 1.1$ and $\Pr(\text{no children}) \approx 0.75\text{--}0.79$, the survival probability per fertile root is small, and the extinction of over 810 type- ≥ 3 roots bounds it at roughly $\lesssim 0.4\%$ in this range.

On the recorded conjecture. The problem thread (via OEIS A229487) records the conjecture that every basin is finite. Theorem 2 *proves* this for 100% of primes $\equiv 3 \pmod{4}$. But the measured $\lambda > 1$ means that, if the full conjecture is true, it is true *near criticality*: extinction is marginal, not comfortable. The honest alternatives are: either $\lambda(x) \rightarrow \leq 1$ (conjecture safe; all basins finite), or $\limsup_x \lambda(x) > 1$ and infinite basins exist but are rare (none below 30,000; per-fertile-root survival $\lesssim 0.4\%$ in range). The present data cannot separate these. We regard the reduction of Q2 to the asymptotics of this Perron root as the main structural insight, and note that it mildly *challenges* the recorded conjecture rather than supporting it.

Conjecture 2. *Every basin has natural density zero. (Proved above for 100% of primes $\equiv 3 \pmod{4}$; open in general. Note this does not follow automatically even if $B(p)$ is infinite.)*

7 What is new here, and what remains open

Already on the thread (not ours): the finiteness/density-zero conjecture (via A229487), and constructed examples with $F = 67$. Apparently new here: Theorem 1(vii) ($F \leq 5\sqrt{n} + 1$); Theorem 2 with its certified exceptional-set verification—a proof of the recorded conjecture for a full-density subfamily of primes, and as far as we can tell the first proven case of Q2/Q3 in any regime; Theorem 3; the exhaustive basin atlas to 30,000 and exhaustive F -records to 10^7 (complementing the thread’s constructed examples); and the multitype-branching reformulation of Q2 with measured near-critical λ . The elementary ingredients of Theorem 2 (the 2-adic valuation of φ ; the classification of $\varphi(n) \equiv 2 \pmod{4}$) are classical; we make no priority claim beyond “apparently unrecorded for this problem.”

Still open: Q2 and Q3 in full (equivalently, whether any tree is infinite; the fate of $p \equiv 1 \pmod{4}$); unconditional unboundedness of F ; any $F(n) \ll n^{o(1)}$ upper bound.

8 Code

`e409.py` (validated φ -sieve, dynamic program for F , exact inverse totient) and `basins.py` (exact basin BFS with certificates). All experiments are reproducible in minutes on a laptop.

References

- [1] T. F. Bloom, *Erdős Problem #409*, <https://www.erdosproblems.com/409>, accessed 2026-07-22.
- [2] L. E. Dickson, A new extension of Dirichlet’s theorem on prime numbers, *Messenger of Mathematics* **33** (1904), 155–161.
- [3] K. Ford, The distribution of totients, *Ramanujan J.* **2** (1998), 67–151.
- [4] R. K. Guy, *Unsolved Problems in Number Theory*, 3rd ed., Springer, 2004. Problem B41.
- [5] OEIS Foundation Inc. (2026), entries A039651 and A229487, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>.